

Questions

Lecture – 1

1. Write down the conclusions of Oersted's Experiment?
2. State the Fleming's Right Hand rule – 1 & 2.
3. Draw the representation of the direction of magnetic field in the plane of paper.
4. Define: Magnetic force, Magnetic flux, 1 Weber, Magnetic field intensity, 1 Tesla.
5. What is the Lorentz force? Write down the vector form of it.
6. An electron is moving with a speed of $5 \times 10^7 \text{ m/s}$ at right angle to a magnetic field of 5000G. (a) What is the magnetic force on the electron? (b) What is the radius of circle in which the electron moves?
7. When an electron moves with a velocity 10^5 ms^{-1} making an angle of 60° with a magnetic field of 0.50T then find the magnitude of the force acting on the electron.
8. When a proton moves with a velocity $2 \times 10^6 \text{ ms}^{-1}$ making an angle of 30° with a magnetic field at a certain place it experiences a force of $4.8 \times 10^{-15} \text{ N}$. Find the magnitude of the magnetic field.
9. When a 10cm long wire carrying 10A current is at an angle of 30° with a magnetic field then it experiences 5N force. Find the magnitude of the magnetic field.
10. At the equator, the earth's magnetic field is nearly horizontal, directed from the southern to northern hemisphere. Its magnitude is about 0.50G. Find the force on a 20m wire carrying a current of 30A parallel to the earth (a) from east to west (b) from north to south.
11. A straight horizontal segment of copper wire carries $i = 28 \text{ A}$. What are the magnitude and direction of the magnetic field needed to float the wire, that is, to balance its weight? Its linear mass density is 46.6g/m.
12. Find the force on each segment of the wire shown in figure below, if $B = 0.15 \text{ T}$. Assume that the current in the wire is 5A.

